Reliability Insights

Large Computational Load Risks Due to Voltage Sensitivity

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Assessing Grid Impacts from Computational Loads

The North American electric grid is projected to experience a sharp rise in the volume of electricity demand. NERC's 2025 *Long-Term Reliability Assessment* (LTRA) projects demand growth at levels higher than at any point in the past two decades. Over the 10-year period covered by the LTRA, aggregated assessment of summer peak demand is projected to rise more than 224 GW. The LTRA also tracks specific demand drivers, and it has shown the significant impact caused by large loads, specifically large computational loads. Computational loads, which include data centers and crypto-mining facilities, have and continue to seek rapid interconnection to the bulk power system (BPS).

Computational loads support artificial intelligence, cloud computing, and related information technology needs that are vital to North America's technological leadership, economic growth, and people's everyday needs. However, the interconnection of large computational loads has implications for BPS reliability, as these loads seek to interconnect very quickly (sometimes within one to two years), creating a risk that the demand will grow faster than available energy. In addition to resource- and energy-adequacy risks, computational loads also introduce challenges to grid operations during normal and event conditions. NERC established the Large Loads Working Group (LLWG) to understand the totality of reliability impacts associated with these large loads and determine any gaps in Reliability Standards and needed mitigations necessary for maintaining the reliability and security of the grid.

Customer-Initiated Load-Reduction Events

One risk detailed by the LLWG and enumerated in two NERC incident reviews (1, 2) is the voltage sensitivity of computational loads. There have been several incidents involving computational loads that have been shown to impact BPS operations when the loads remove themselves from the system during system disturbances. Several voltage-sensitive customer-initiated load reductions have occurred in the Eastern Interconnection and the Electric Reliability Council of Texas (ERCOT) over the past two years.

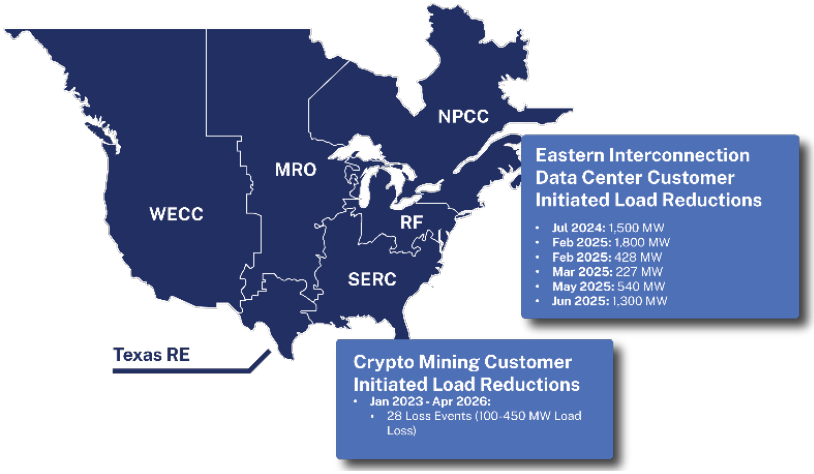


Figure 1: Customer-Initiated Load Reduction Events

The incidents largely unfold in a familiar pattern:

- The grid experiences a normally cleared fault, and voltage dips occur as part of normal reclosing to isolate the fault.
- After several reclosing attempts, computational load entities react to the voltage dips by transferring their load to their backup power systems, removing themselves from the grid, with the goal of protecting their equipment from outages and any potential damage resulting from prolonged voltage dips.
- As a result, large amounts of demand suddenly disappear from the BPS, creating oscillations on the systems as generation rebalances frequency and voltage to meet the remaining demand.

Historically, the electric grid has not experienced, nor planned or operated for, these significant amounts of simultaneous load losses in response to a fault on the system. Rather, planners and operators have traditionally focused on the risk of large generation losses (and generators must meet certain Reliability Standards associated with "riding through" these types of faults). While these recent load reduction incidents did not create any near- or long-term issues for the system, future incidents could result in severe impacts given the expansion in the number and size of computational load sites, especially when their deployment is geographically concentrated (such as in "Data Center Alley" in Virginia). Additionally, since all computational loads on a circuit could see the same disturbance and take the same action, this creates significant common-mode failure risk.

Corrective Actions Can Help Mitigate Impacts

Recently, large computational load entities have worked with grid operators to reduce the likelihood of customer-initiated load reduction and improve ride-through performance. Based on discussions with these operators, certain data centers adjusted the protection and control schemes that they use to determine when their facilities transfer their load to a backup power supply. Specifically, they lengthened the time duration required for measuring voltage dips to be "counted" and altered the number of depressions during a specified period that would trigger disconnection from the grid. The transmission entity also included these requirements for all newly connected facilities.

On April 18, 2026, a transmission system fault occurred in the same vicinity and with similar characteristics to an event in July 2024 that had led to 1,500 MW of load loss. Yet, this April 2026 event saw only 500 MW of sustained data center load loss. The improved performance may be attributable to the changes made by the data centers, and it demonstrates how collaboration can help mitigate risks to BPS reliability.

While such actions have been important and demonstrate the technical feasibility of computational load ride-through, computational loads still present challenges to grid reliability. Most recently, in July 2026, a nearly 4 GW customer-initiated load reduction occurred in Virgina, demonstrating that the risk is increasing, not abating. The specifics of this incident are still being analyzed as of September 2026. This ongoing risk underscores the need for consistent and uniform best practices throughout North America to ensure the reliability, resilience, and security of the BPS as more computational loads are interconnected.

NERC Actions to Address Computational Loads

NERC has taken action to understand the effects of computational large loads on the BPS and identify effective, efficient, and reliable pathways for their integration. In addition to establishing the LLWG, NERC developed a holistic Large Loads Action Plan, which involves efforts such as publishing technical resources and alerts and conducting extensive engagement and outreach to computational load entities. NERC has determined through these efforts that additional action is required to address large computational loads.

Key among these actions is to modernize NERC's reliability approach and bring large computational loads into the reliability framework. To that end, NERC is developing registry criteria, based on specific physical and electrical criteria, that will define which entities will be required to register with NERC and to help develop and comply with Reliability Standards. The Regional Entities, working in close coordination with NERC, are ready to assist in the registration process once the registry criteria are final.

NERC is also developing Reliability Standards for existing and newly registered entities, starting with foundational reliability activities, such as commissioning, communications, and modeling. The Federal Energy Regulatory Commission has directed NERC to complete the registry criteria and foundational Reliability Standards by the end of 2026. NERC will consider additional Reliability Standards moving forward to address specific performance requirements such as ride-through, modeling requirements, and critical cyber and physical security protections, including supply-chain risk management.

The North American BPS has reached a historic juncture defined by the rapid change in the nature and volume of electricity demand. Recent customer-initiated load reduction events underscore the potential significant risks that large computational loads may present to BPS reliability. Fortunately, mitigation options are available, and computational load stakeholders have taken proactive steps to work with NERC, the Regional Entities, and grid operators.

While much work remains, this collaborative approach is an example of how stakeholders can engage with grid authorities to find mutually beneficial solutions. NERC and the Regional Entities seek to expand and deepen collaboration with computational load stakeholders as this work continues to support the reliable operation of the BPS.